

Application of Information & Communication Technologies

Types of Networks

Lecture-21

Recap of Lecture 20

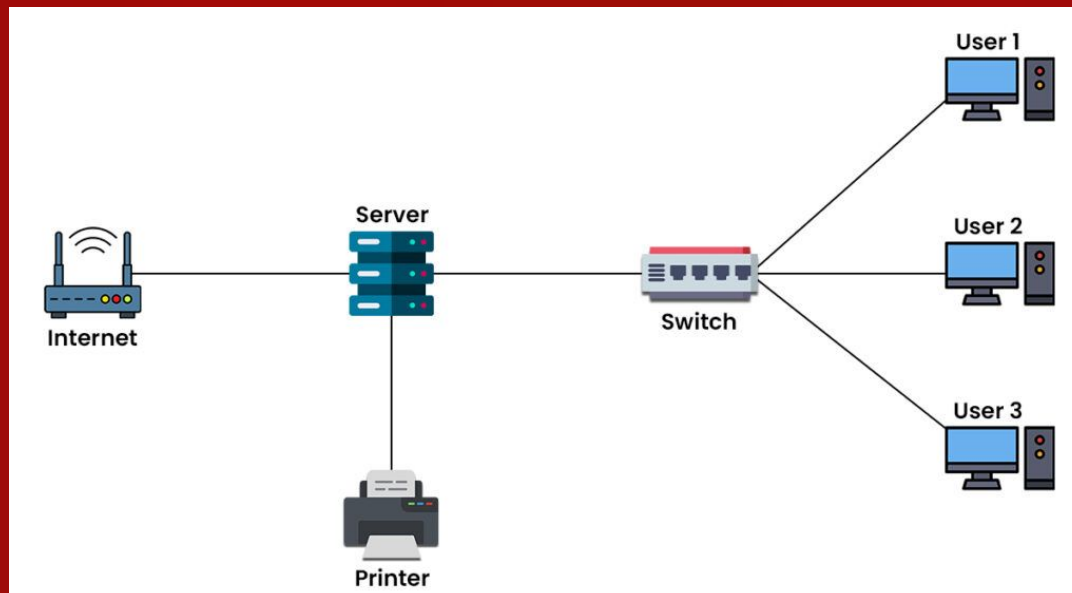
- ◆ Computer communication Concepts:
 - Meaning and Importance of Computer Communication
 - Components of Communication System
 - Sending Device
 - Channel
 - Receiving Device

Overview of Lecture 21

- ◆ Define what a computer network is.
- ◆ Describe different types of networks (LAN, MAN, WAN).
- ◆ Understand wireless networks (WLAN, WWAN).
- ◆ Identify examples of networks used in daily life.

Computer Network

- A Network is simply two or more computers that are linked together.
- A network is a group of connected computers and devices that share data and resources.
- Purpose: Share files, internet, printers, and applications.

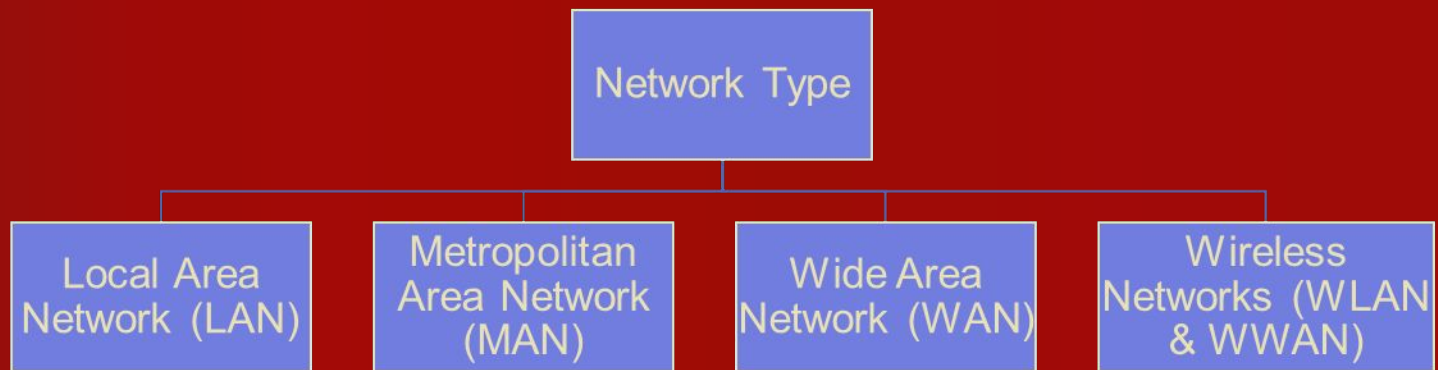


Why Networking?

- ◆ Enables **communication** between users.
- ◆ Facilitates **data sharing** and resource utilization.
- ◆ Reduces **cost** by sharing devices.
- ◆ Enhances **collaboration** and remote work.

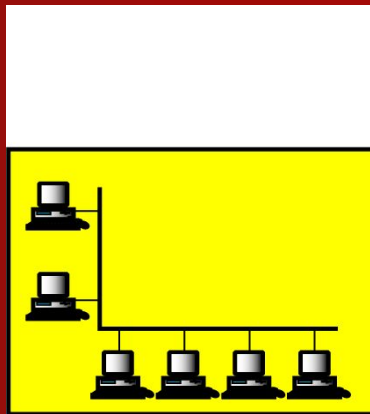
Classification of Networks

- ◆ Networks can be categorized by size and coverage area:

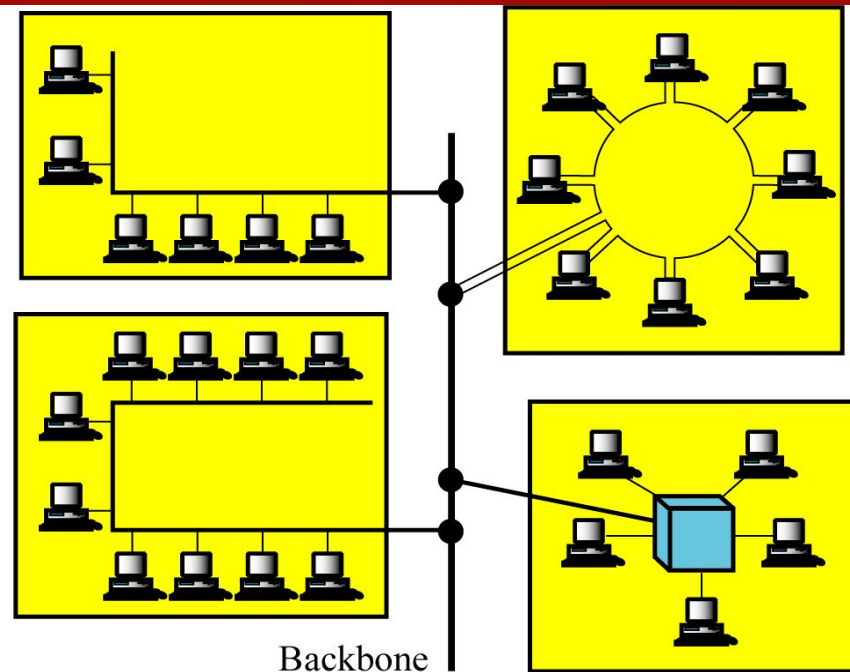


Local Area Networks

- ◆ Covers a **small area** such as a room, building, or campus.
- ◆ **High speed** (up to 1 Gbps or more).
- ◆ **Privately owned** and maintained.
- ◆ Common in **schools, offices, and labs.**



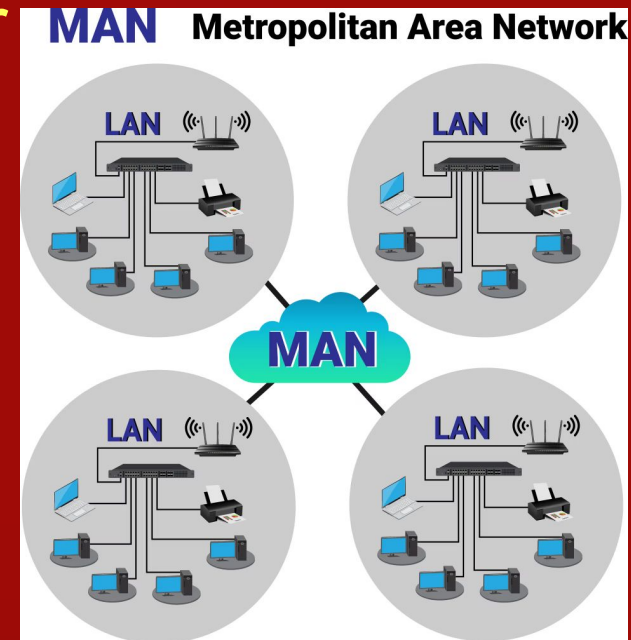
a. Single building LAN



b. Multiple building LAN

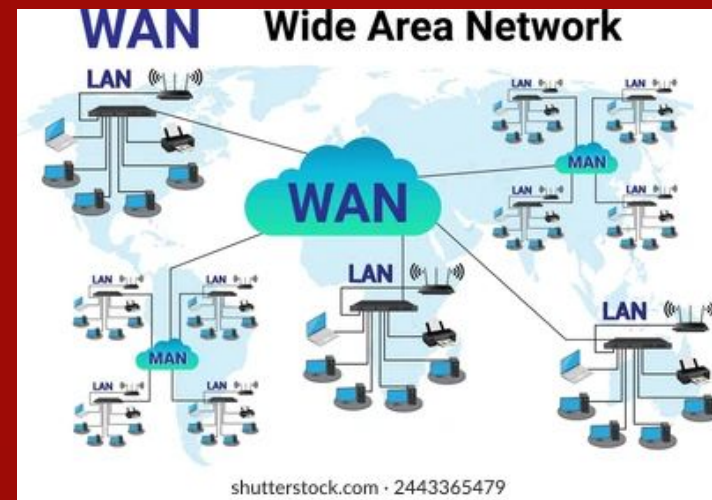
Metropolitan Area Network (MAN)

- ◆ Covers a city or large campus.
- ◆ Connects multiple LANs using fiber optics or wireless.
- ◆ Managed by ISPs or city authorities.
- ◆ Example: city-wide Wi-Fi or university campus network.



Wide Area Network (WAN)

- ◆ Covers large geographical areas (countries or continents).
- ◆ Connects multiple MANs or LANs.
- ◆ Uses telecommunication lines, satellites, or undersea cables.
- ◆ Example: the Internet is the largest WAN.



Wireless Networks

- ◆ **WLAN (Wireless LAN):**
- ◆ Covers a small area using Wi-Fi.
- ◆ Common in homes, offices, cafés.
- ◆ **WWAN (Wireless WAN):**
- ◆ Covers wide areas using cellular networks (4G/5G).
- ◆ Used for mobile data and long-distance communication.



Advantages of Wireless Networks

- ◆ **Mobility:** users can move freely within the network.
- ◆ **Easy installation:** no cables needed.
- ◆ **Flexibility:** connects many devices (IoT, smartphones).
- ◆ **Scalability:** easy to expand coverage.

Examples of Networks in Daily Life

Scenario	Network Type	Purpose
Home Wi-Fi	WLAN	Internet access, smart devices
Office LAN	LAN	File sharing, printer sharing
Bank ATMs	WAN	Central database connection
University network	MAN	Inter-campus connectivity
Mobile Internet	WWAN	4G/5G data services

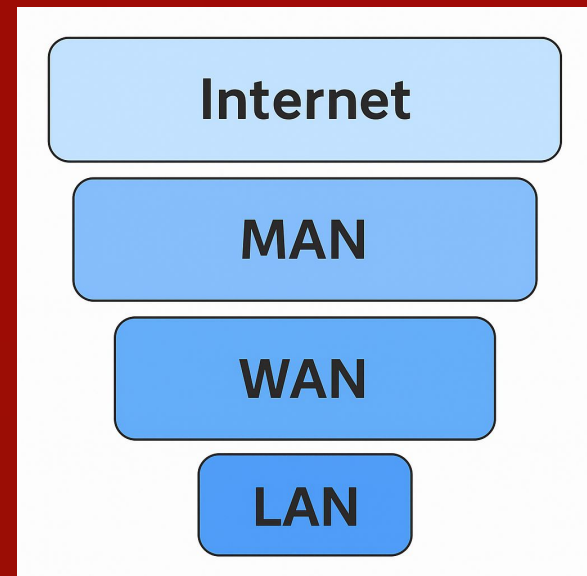


Comparison Summary Table

Network Type	Full Form	Coverage Area	Speed	Example
LAN	Local Area Network	Building / Campus	High	Office, School
MAN	Metropolitan Area Network	City / Campus	Medium	University Network
WAN	Wide Area Network	Country / Globe	Variable	Internet
WLAN	Wireless Local Area Network	Room / Office	High	Wi-Fi Network
WWAN	Wireless Wide Area Network	Large Region	Medium	4G/5G Mobile Data

Summary

- **LAN** – small, fast, local.
- **MAN** – connects LANs in a city.
- **WAN** – connects cities and countries.
- **WLAN/WWAN** – wireless access using radio signals.
- Networks are essential for **communication and resource sharing** worldwide.



Suggested Reading

- ◆ Ch-07, “Understanding Computers: Today and Tomorrow, Comprehensive”, 15th Edition by Deborah Morley & Charles S. Parker
- ◆ Ch-08, Discovering Computers Fundamentals-Your Interactive Guide -- Gary B Shelly; Misty E Vermaat; Jeffrey J Q